Lecture 20 2025-11-06

Idea: $\int_a^b f(x)\,dx = \sum_{i=0}^{n} a_i f(x_i) + \text{error}$

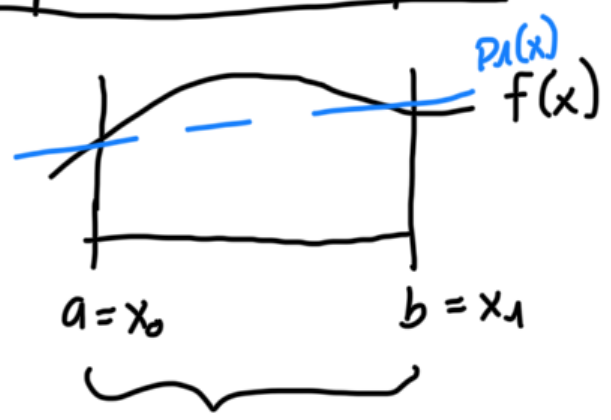
Approximate f by a polynomial $P_n(x)$ and then

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx = \sum_{i=0}^{n} f(x_i) \underbrace{\int_a^b L_i(x)\,dx}_{:=a_i \text{ weights}}$$

Error: $E(f) := \int_a^b (f(x) - P_n(x))\,dx = \int_a^b u(x) \prod_{k=0}^{n} (x - x_k)\,dx$

$$u(x) = f[x_0, \dots, x_n, x] = \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \qquad \xi(x) \in [a,b]$$

(Important) Examples: * **Trapezoidal rule** $(n=1)$



(approximate f by linear polynomial)

$$\int_a^b f(x)\,dx = f(x_0)\int_a^b L_0(x)\,dx + f(x_1)\int_a^b L_1(x)\,dx + E(f)$$

$$= \frac{b-a}{2}\left(f(x_0) + f(x_1)\right) + E(f)$$

$h = b - a$

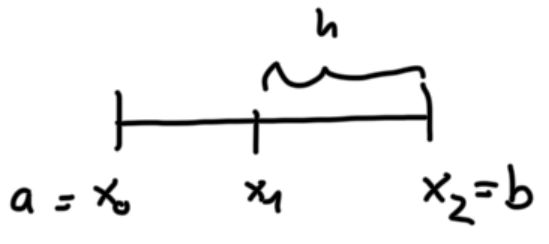
$$E(f) = u(\xi)\int_a^b \underbrace{(x-a)(x-b)}_{\leq 0 \text{ on } [a,b]}\,dx \qquad \text{MVT for integrals}$$

$$\int_a^b (x-a)(x-b)\,dx = -\frac{(b-a)^3}{6} = -\frac{h^3}{6}$$

$$\Rightarrow E(f) = -\frac{h^3}{6}u(\xi) = -\frac{h^3}{12}f''(\tilde{\xi})$$

Simpson's rule $(n=2)$

approximate f by P_2



$$h = \Delta x_0 = \Delta x_1 = \frac{b-a}{2}$$

$$\Rightarrow \int_a^b f(x)\,dx = \frac{h}{3}\left(f(x_0) + 4f(x_1) + f(x_2)\right) - \frac{h^5}{90} f^{(4)}(\xi)$$

$x_0 = a,\ x_1 = \frac{a+b}{2},\ x_2 = b$

(evaluate $\int_a^b L_i(x)\,dx$ for quadratic polynomials)

$n = 3$ & $n = 4$ rules see textbook p. 197 (10th edition, Sec. 4.3)

These are called <u>Closed Newton-Cotes Formulas</u> because the endpoints a and b are included as nodes, $a = x_0$, $b = x_n$

<u>Error formulas:</u>

<u>n even:</u> $E(f) = \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!} \int_0^n t^2(t-1)\cdots(t-n)\,dt$

<u>n odd:</u> $E(f) = \frac{h^{n+2} f^{(n+1)}(\xi)}{(n+1)!} \int_0^n t(t-1)\cdots(t-n)\,dt$

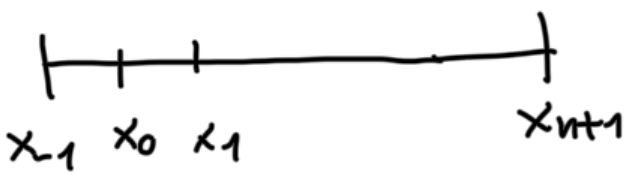
integral transformation $x_j = x_0 + jh$

Hard to prove for $n \geq 3$ since $t(t-1)\cdots(t-n)$ is not of constant sign, so we cannot use the MVT for integrals (see Hayes/Rubin 1970 The American Mathematical Monthly)

Open Newton-Cotes formulas

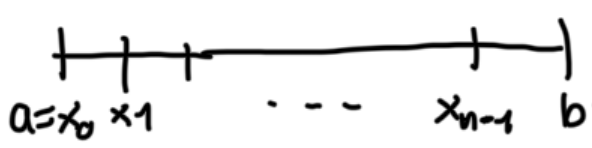
- Do not include endpoints as quadrature points, only use interior nodes

<u>book:</u>



confusing?

we use instead:



$h = x_j - x_{j-1}$

$x_1, \ldots, x_{n-1}$ quadrature points

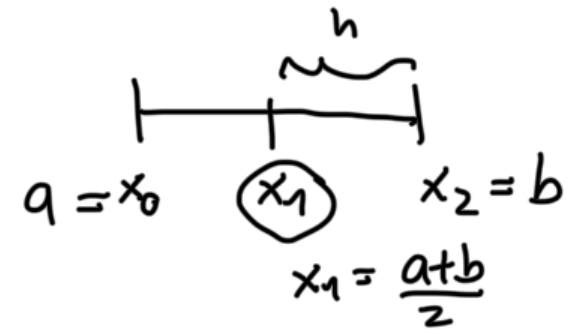
so formulas will involve $f(x_1), \ldots, f(x_{n-1})$

<u>Now:</u> $\int_a^b f(x)\,dx \approx \sum_{i=1}^{n-1} a_i f(x_i)$ $\quad a_i = \int_a^b L_i(x)\,dx$ (deg $= n-2$)

zero at $x_1, \ldots, x_{n-1}$ except $L_i(x_i) = 1$

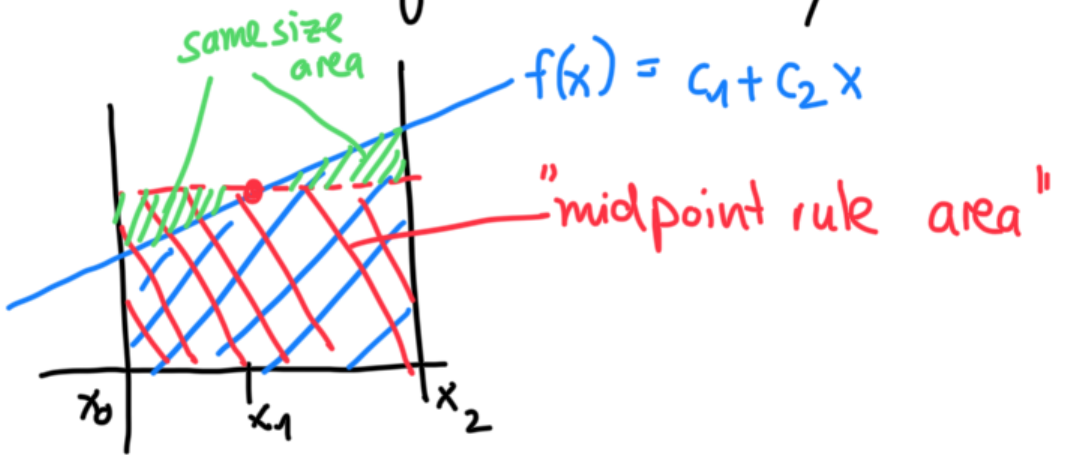
Most important open Newton-Cotes formula:

Midpoint rule $n=2$



$$\int_a^b f(x)\,dx = \underbrace{2h\, f(x_1)}_{a_1 = \int_0^{2h} L_1(x)\,dx} + \underbrace{\frac{h^3}{3} f''(\xi)}_{\hookrightarrow \frac{h^3}{2!}\int_0^2 (t-1)^2\,dt}$$

As with Simpson's rule, we gain accuracy in the even case. (symmetry)



General error formulas

n even: $$E(f) = \frac{h^{n+1} f^{(n)}(\xi)}{n!} \int_0^n (t-1)^2 (t-2)\cdots(t-n+1)\,dt$$

n odd: $$E(f) = \frac{h^n f^{(n-1)}(\xi)}{(n-1)!} \int_0^n (t-1)\cdots(t-n+1)\,dt$$

Def: Degree of precision: Largest m such that $\int_a^b p(x)\,dx = \sum_{j=0}^{n} a_j p(x_j)$ for all polynomials p of deg $\le m$

(open Newton-Cotes $n-1$; $\hookrightarrow j=1$ open Newton-Cotes)

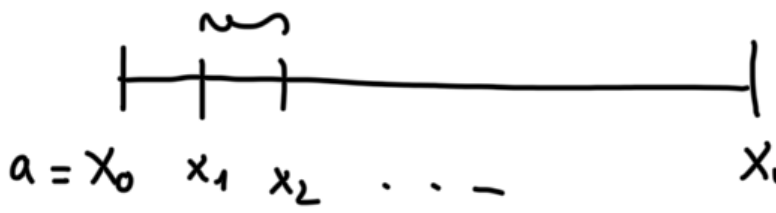
trapezoidal rule $m=1$

Simpson's rule $m=3$

Midpoint rule $m=1$

4.4. Composite Numerical Integration

Use piecewise polynomial interpolation instead of one higher-order polynomial



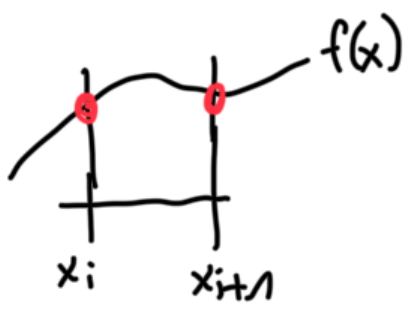
$$h = \Delta x_i = \frac{b-a}{n}$$

For example:

Composite trapezoidal rule: Linear polynomial in each $[x_i, x_{i+1}]$ to approximate f

$$\int_a^b f(x)dx = T(h) - \frac{(b-a)}{12} h^2 f''(\mu) \quad \text{for } \mu \in (a,b)$$

$$T(h) = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)\right]$$



$$\int_{x_i}^{x_{i+1}} f(x)dx \approx \frac{h}{2}\left(f(x_i) + f(x_{i+1})\right)$$

→ sum over i to get $T(h)$

The error comes from summing the errors on each subinterval

$$E_1 = -\frac{h^3}{12} f''(\xi_1) \quad \xi_1 \in (x_0, x_1)$$

$$E_2 = -\frac{h^3}{12} f''(\xi_2) \quad \dots$$

$$\Rightarrow E = E_1 + E_2 + \dots + E_n = -\frac{h^2}{12}\left(h f''(\xi_1) + \dots + h f''(\xi_n)\right)$$

$$= -\frac{h^2}{12}\left(\underbrace{hn}_{b-a} \underbrace{\frac{f''(\xi_1) + \dots + f''(\xi_n)}{n}}\right)$$

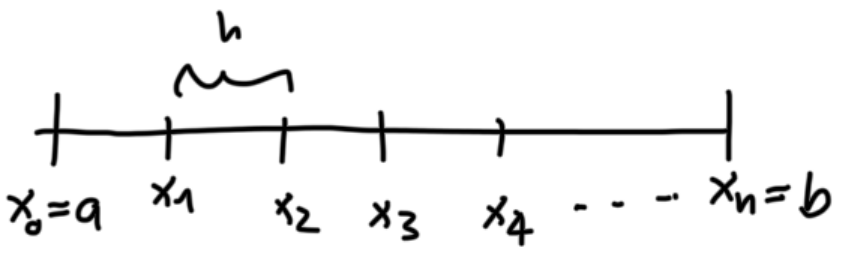
(*) average value is between $\min_j f''(\xi_j)$ and $\max_j f''(\xi_j)$

By IVT, $\exists \mu$ between $\xi_{\min}$ (at $\operatorname{argmin}_j f''(\xi_j)$) and $\xi_{\max}$ (at $\operatorname{argmax}_j f''(\xi_j)$) s.t. $f''(\mu) = \frac{f''(\xi_1) + \dots + f''(\xi_n)}{n}$

$$\Rightarrow E = -\frac{b-a}{12} h^2 f''(\mu)$$

Composite Simpson's Rule

$$\int_a^b f(x)dx = S(h) - \frac{b-a}{180} h^4 f^{(4)}(\mu)$$



where $S(h) = \frac{h}{3}\left(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots \pm 2f(x_{n-2}) + 4f(x_{n-1})\right.$ (n even)

$+ f(x_n)\Big)$

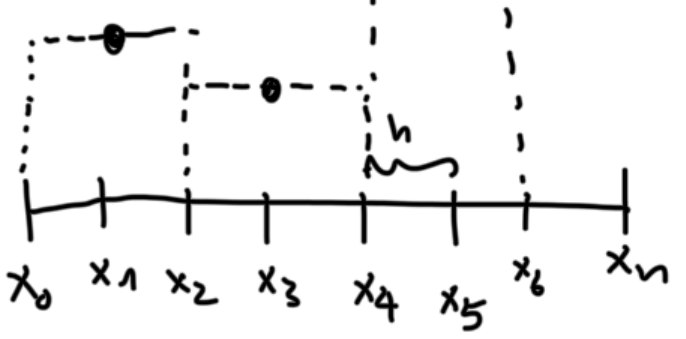
in each $[x_i, x_{i+2}]$ we approximate $\int_{x_i}^{x_{i+2}} f(x)\,dx \approx \frac{h}{3}\left(f(x_i)+4f(x_{i+1})+f(x_{i+2})\right)$

and then sum

$$E_i = -\frac{h^5}{90} f^{(4)}(\xi_i)$$

$$E(f) = E_1 + E_2 + \ldots + E_{n/2} = -\frac{h^4}{90}\underbrace{h\frac{n}{2}}_{=\frac{b-a}{2}}\underbrace{\left(\frac{f^{(4)}(\xi_1)+\ldots+f^{(4)}(\xi_{\frac{n}{2}})}{n/2}\right)}_{|VT| := f^{(4)}(\mu)\quad \mu\in(a,b)}$$

Composite Midpoint rule



$$\int_a^b f(x)\,dx = M(h) + \frac{b-a}{6}h^2 f''(\mu) \quad \mu\in(a,b)$$

$$M(h) = 2h\sum_{j=1}^{n/2} f(x_{2j-1})$$

(apply midpoint rule in each $[x_{2i}, x_{2i+2}]$

$i = 0, \ldots \frac{n}{2}-1$)